

Netflix Recommendation System using Collaborative Filtering and Matrix Factorization

Technical Report

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1 Problem Understanding

Recommendation systems have become a critical component of modern digital platforms, helping users discover relevant content while improving engagement and retention. Streaming platforms such as Netflix rely heavily on recommendation algorithms to personalize content delivery and enhance user experience.

The objective of this project is to develop and evaluate personalized movie recommendation systems using the Netflix Prize Dataset. The system should be capable of:

- Learning user preferences from historical interactions
- Predicting ratings for unseen movies
- Generating personalized movie recommendations
- Comparing multiple recommendation approaches
- Evaluating recommendation quality using ranking and prediction metrics

To achieve these objectives, three approaches were explored:

1. Popularity-Based Recommendation
2. Item-Based Collaborative Filtering
3. Singular Value Decomposition (SVD) Matrix Factorization

2 Dataset Description

Dataset Overview

The project utilizes the Netflix Prize Dataset, one of the most influential datasets in recommendation system research.

Metric	Value
Ratings	100,480,507
Users	480,189
Movies	17,770
Rating Scale	1–5
Additional Data	Movie Title, Release Year

Table 1: Original Dataset Statistics

Due to computational limitations, a filtered subset was used for model development.

Metric	Value
Ratings	18,687,764
Users	148,083
Movies	4,491

Table 2: Filtered Dataset Statistics

Data Preprocessing

The following preprocessing steps were applied:

- Parsed Netflix Prize text files into structured tabular format
- Converted data types to memory-efficient formats
- Merged movie metadata with ratings
- Converted rating dates to datetime format
- Filtered movies with insufficient ratings
- Filtered inactive users
- Generated a clean user-item interaction dataset

3 Exploratory Data Analysis

Rating Distribution

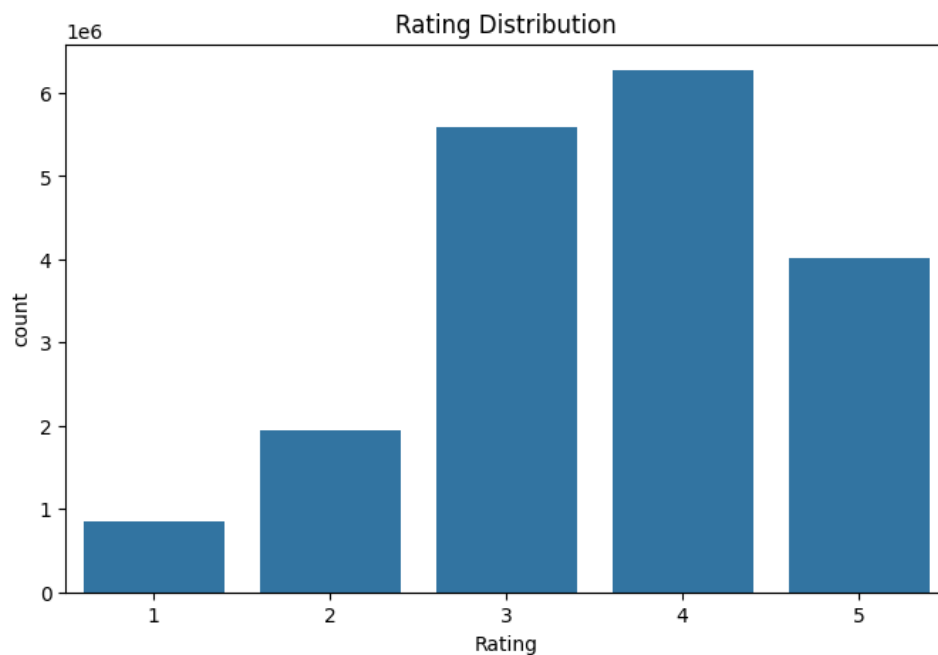


Figure 1: Rating Distribution

Observations:

- Ratings are skewed toward positive values.
- Ratings of 3, 4, and 5 dominate the dataset.
- Users generally provide favorable feedback.

Business Implication: Positive rating bias suggests that recommendation systems should focus on ranking relevant content rather than merely predicting rating values.

Most Popular Movies

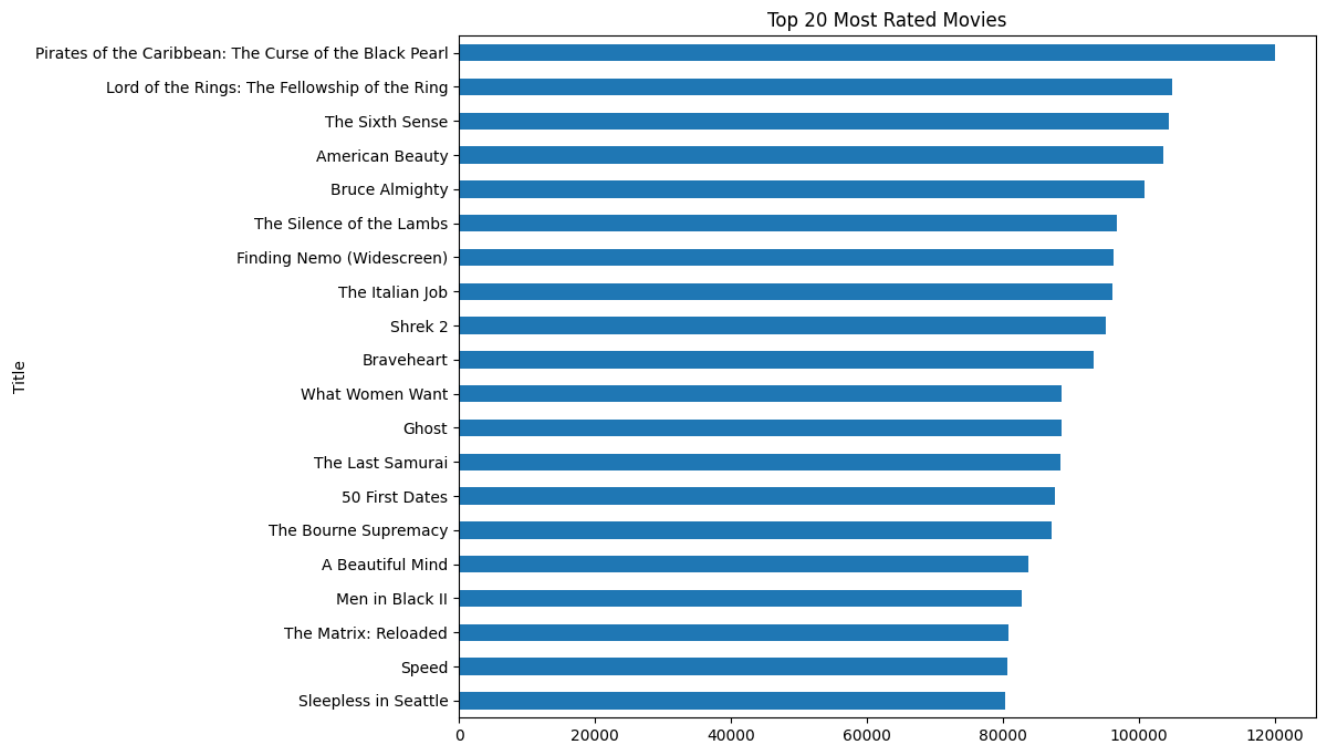


Figure 2: Most Popular Movies

Observations:

- A small subset of movies accounts for a significant portion of ratings.
- Movie popularity follows a long-tail distribution.

Business Implication: Popularity-based recommendation systems can effectively recommend mainstream content but struggle with personalization.

User Activity Distribution

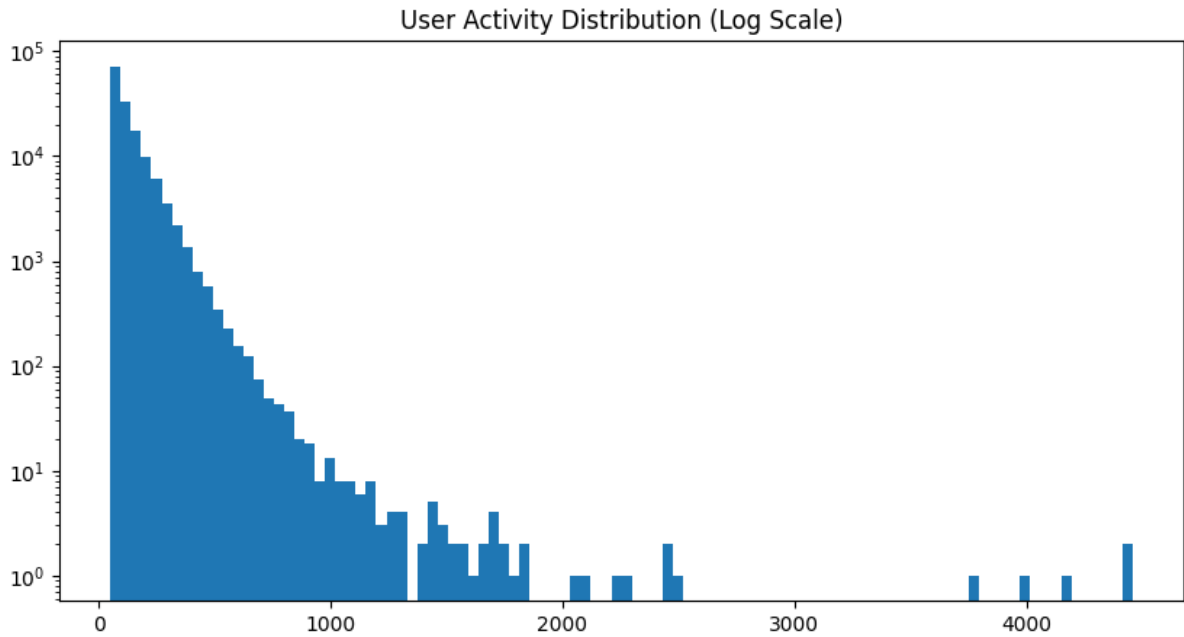


Figure 3: Distribution of User Activity

Observations:

- User activity is highly skewed, with most users rating only a small number of movies.
- A small group of highly active users contributes a large share of the ratings.
- This long-tail behavior is common in large-scale recommendation systems.

Business Implication: Highly active users provide valuable interaction data that helps recommendation models learn user preferences more effectively, while less active users make personalization more challenging.

Sparsity Analysis

The user-item matrix is extremely sparse.

Metric	Value
Sparsity	97.19%
Density	2.81%

Table 3: Dataset Sparsity Analysis

Interpretation: Most users interact with only a tiny fraction of available movies. This creates a challenging recommendation environment and motivates the use of collaborative filtering and latent factor models.

4 Methodology

4.1 Popularity-Based Recommendation

The popularity model recommends movies based on:

- Average rating
- Number of ratings

Advantages:

- Simple implementation
- Fast computation
- Effective for new users

Limitations:

- No personalization
- Recommends the same content to all users

4.2 Item-Based Collaborative Filtering

Item-Based Collaborative Filtering identifies relationships between movies based on historical rating behavior.

Process:

- Construct user-item interaction matrix
- Compute item similarity
- Identify similar movies
- Generate personalized recommendations

Advantages:

- Personalized recommendations
- Easy to interpret

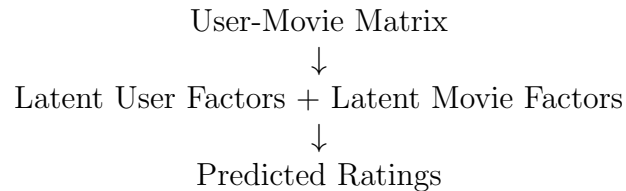
Limitations:

- Struggles with sparse data
- Computationally expensive at scale

4.3 Matrix Factorization (SVD)

Singular Value Decomposition (SVD) decomposes the user-item interaction matrix into latent user and movie factors.

Workflow:



Advantages:

- Captures hidden preference patterns
- Handles sparse datasets effectively
- State-of-the-art classical recommendation approach

Limitations:

- Less interpretable than neighborhood methods
- Requires model training

5 Model Design

Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

Evaluation Metrics

Three evaluation metrics were used:

- **RMSE (Root Mean Squared Error)**
 - Measures rating prediction accuracy.
 - Lower values indicate better performance.
- **MAP@10**
 - Measures recommendation ranking quality.
 - A movie was considered relevant if: $\text{Rating} \geq 3.5$
 - Higher values indicate better ranking performance.
- **Precision@10**
 - Measures the proportion of relevant items within the top 10 recommendations.
 - Higher values indicate better recommendation quality.

6 Experimental Results

Model	RMSE
Item-Based CF	1.0429
SVD	0.9631

Table 4: RMSE Results

Model	MAP@10
Item-Based CF	0.8307
SVD	0.8792

Table 5: MAP@10 Results

Model	Precision@10
Item-Based CF	0.1698
SVD	0.1703

Table 6: Precision@10 Results

Observations:

- SVD achieved the lowest RMSE.
- SVD achieved the highest MAP@10.
- Matrix Factorization outperformed Item-Based Collaborative Filtering across both evaluation metrics.
- A similar trend was observed for Precision@10, where SVD achieved the highest recommendation relevance among the evaluated models.

7 Recommendation Examples

Selected User

User ID: 305344

User Preference Profile

Movie Title	Rating
Yanni: Live at the Acropolis	5
18 Again	5
Yellow Submarine	5
The Return of the Living Dead	5
The Passion of the Christ	5
The People vs. Larry Flynt	5
The Perfect Storm	5
Jefferson Starship: The Definitive Concert	5
Lord of the Rings: The Fellowship of the Ring	5
Shiloh 2: Shiloh Season	5

Table 7: Top Rated Movies

Recommended Movies Generated by SVD

Movie	Predicted Rating
Doctor Zhivago	3.5082
Nuremberg	2.8792
Ray	2.8545
The Incredibles: Bonus Material	2.6354
Hackers	2.3565
The Rookie	2.0497
Sideways	2.0356
The Pledge	1.9433

Table 8: SVD Recommendations

Recommendation Analysis

The recommendations demonstrate the ability of the SVD model to identify unseen content based on latent user preferences rather than simple popularity trends. The model successfully leverages collaborative information from similar users and movies to generate personalized recommendations.

8 Success Cases and Failure Cases

Success Cases

- Personalized recommendations generated successfully
- Strong ranking quality achieved
- SVD captured latent user preferences effectively
- Better performance than neighborhood-based methods

Failure Cases

- New users with no rating history
- New movies with limited interactions
- Preference drift over time
- Highly sparse interaction patterns

9 Key Insights

- Dataset sparsity remains one of the primary challenges in recommendation systems.
- Popularity-based methods provide strong baseline recommendations but lack personalization.
- Item-Based Collaborative Filtering improves personalization but suffers from scalability and sparsity issues.
- SVD significantly outperformed Item-Based Collaborative Filtering in both prediction accuracy and ranking quality.
- Ranking metrics such as MAP@10 provide more meaningful evaluation than RMSE alone when assessing recommendation usefulness.

10 Future Improvements

Potential enhancements include:

- **Neural Collaborative Filtering:** Deep learning models capable of learning complex user-item interactions.
- **Hybrid Recommendation Systems:** Combining collaborative filtering with content-based information.
- **Explainable Recommendations:** Providing transparent explanations for recommendations.
- **Temporal Modeling:** Capturing changes in user preferences over time.
- **Real-Time Recommendation Pipelines:** Generating recommendations dynamically in production environments.

Conclusion

This project developed and evaluated multiple recommendation approaches using the Netflix Prize Dataset. The results demonstrated that Matrix Factorization (SVD) consistently achieved superior performance across rating prediction, recommendation ranking, and recommendation relevance metrics.

- $\text{RMSE} = 0.9631$
- $\text{MAP@10} = 0.8792$
- $\text{Precision@10} = 0.1703$

Compared to Item-Based Collaborative Filtering, SVD provided both more accurate rating predictions and higher-quality recommendation rankings. The findings highlight the effectiveness of latent factor models for large-scale recommendation systems operating in sparse user-item environments.